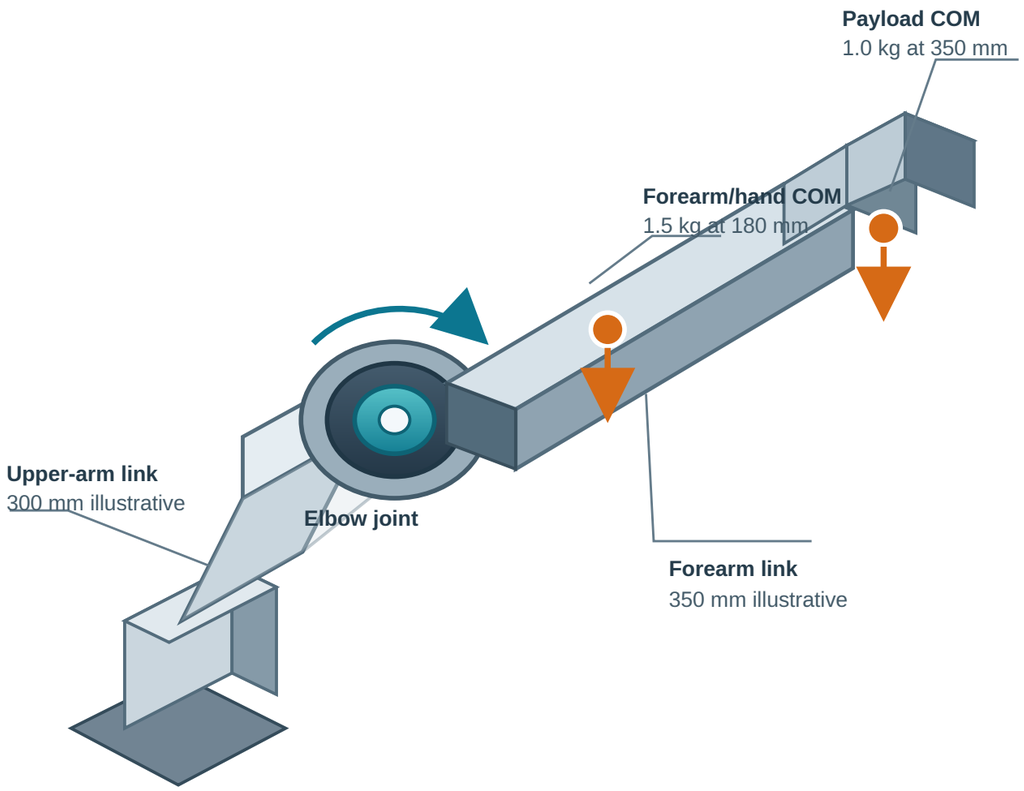
K-EL/01 | Split-Yoke Elbow Actuator

Kritva Robotics KR-MECH-INT-001 | Concept artwork and assembly/load-path illustration | Not for manufacture

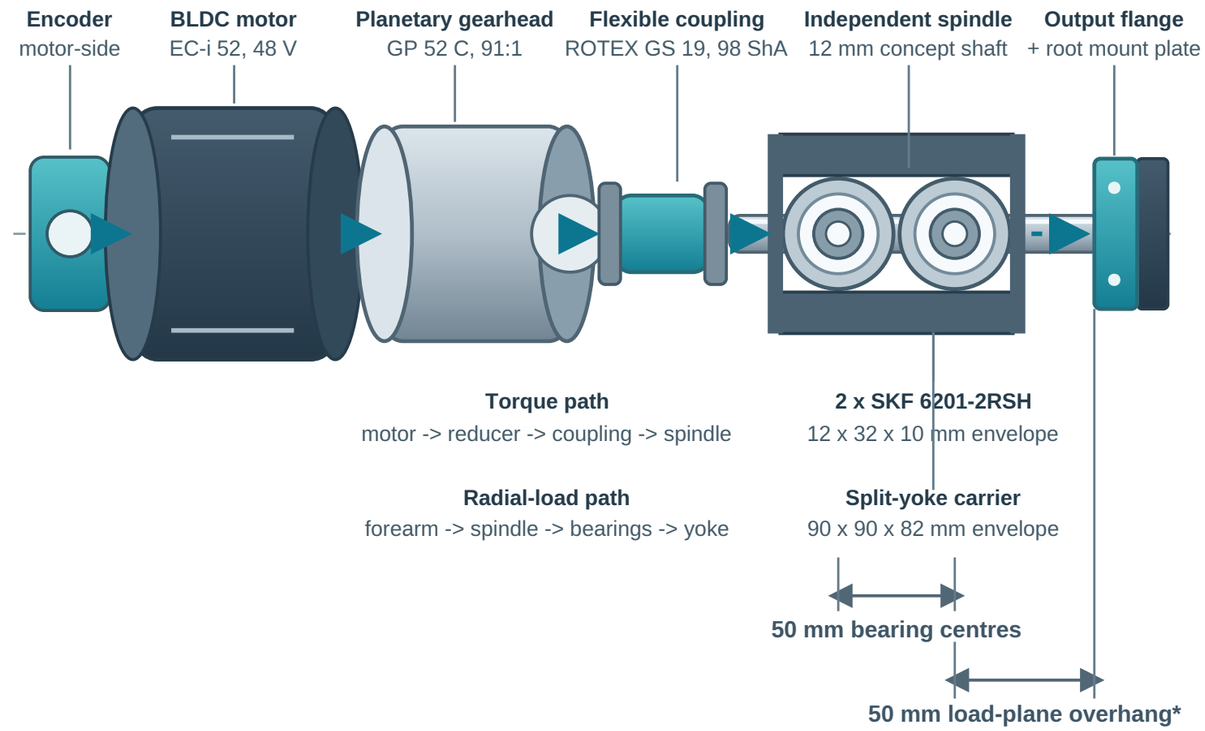
CONCEPT ONLY - DESIGN GATES OPEN

A. Simplified robotic arm - isometric concept



Load path: forearm -> spindle + bearings -> split-yoke -> upper arm

B. Elbow actuator - exploded side concept



*From rear bearing centre (Y=61 mm) to forearm link center plane (Y=111 mm) in this envelope model.

C. Design-point calculations and explicit trade-off

Static gravity torque	Preliminary output target	Speed and reduction	Backlash trade-off	Mechanical concept assumptions
6.082 N m	**12.164 N m**	**45 rpm / 91:1**	**1 degree average**	**12 mm spindle | 50 mm bearing spacing**
1.0 kg x 350 mm + 1.5 kg x 180 mm	2.0 x static; assumed margin only	Motor speed screen = 4,095 rpm	At 350 mm radius: about 6.1 mm endpoint arc	**50 mm axial overhang | key 4 x 4 x 20**
				All are envelope values; detail design remains open.

Key unresolved gates: confirmed maxon motor/gearhead/encoder configuration and loaded efficiency; dynamic duty and thermal performance; coupling hub fit/service factor; shaft fatigue/deflection and bearing life; accuracy and safe power-loss behavior.

Source requirements: Kritva Robotics KR-MECH-INT-001. Component values are preliminary catalogue references. The 2x torque target is not a certified safety factor. No purchase, manufacture, safety certification, FEA, or physical test is claimed.